# Parallel circuits



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Relate branch currents to a shared potential difference.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 7, Day 5; 3 course days later.*
Q=It=15 C.

**02. Explain a worked example (worked)**
Same 6 V across branches; 1+2=3 A at each junction.

**03. Look and explain (model)**
Yes, if its own complete path to the ideal supply remains.

**04. Build the solution (practice)**
2 A and 1 A; total 3 A.

**05. Use the idea (reason)**
Adding a conducting path increases total current for the same voltage; $R_{eq}=V/I_{total}$ decreases.

**06. Choose a method (mixed)** *Review: Unit 6, Day 14; 12 course days later.*
A diverging lens spreads initially parallel rays outward.

**07. Exit check (exit)**
.5 A.

**08. STRETCH: Apply it another way (stretch)**
Existing currents unchanged; new branch 1 A; total 4 A, assuming ideal maintained voltage.

# Parallel circuits



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**Learning target:** Relate branch currents to a shared potential difference.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 7, Day 5; 3 course days later.*
Circuit response involves propagation of electromagnetic changes and local charge motion. Drift speed is not the propagation speed of the signal.

**02. Explain a worked example (worked)**
Same 6 V across branches; 1+2=3 A at each junction.

**03. Represent and interpret (model)**
Yes, if its own complete path to the ideal supply remains.

**04. Solve and justify (practice)**
2 A, 1 A; total 3 A; $R_{eq}$=4 Ω.

**05. Analyze the reasoning (reason)**
Adding a conducting path increases total current for the same voltage; $R_{eq}=V/I_{total}$ decreases.

**06. Choose a method (mixed)** *Review: Unit 6, Day 14; 12 course days later.*
$d_i \approx -13.33$ cm; $m \approx +1.667$. Upright, enlarged virtual image.

**07. Exit check (exit)**
4 Ω parallel; 16 Ω series.

**08. STRETCH: Apply it another way (stretch)**
$R_{eq}$=2.4 ohms; I=5 A; power 60 W, also 36+24 W.